src/dynamics.py · gravity_vector MMA3001 DeepONet project

gravity_vector

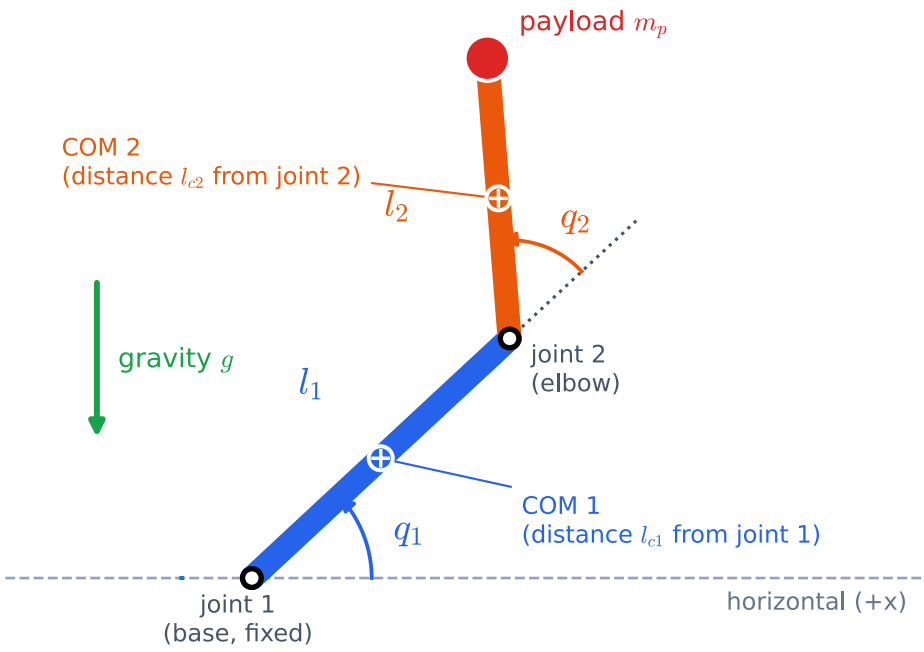
The torque needed to hold the arm still against gravity, derived from potential energy and checked with lever arms

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The set-up: a two-link robot arm in a vertical plane

Everything in this project describes one physical object: a planar robot arm with two rigid links, two motorised joints, bolted to a wall at the origin. It moves in a *vertical* plane, so gravity matters. At the tip of link 2 we may clip on an extra weight (the **payload**). Each joint has a little friction that slows it down in proportion to how fast it spins.



The arm. Blue = link 1, orange = link 2. The angle q_1 is measured from the horizontal; the angle q_2 is measured from the *straight-on extension* of link 1 (dotted), not from the horizontal. The dots with a white cross are the links' centres of mass (COM).

Conventions (fixed everywhere in the project): the state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$; gravity points in the $-y$ direction; potential energy is zero at the height of joint 1; positive angles and torques are anticlockwise.

Symbol	Meaning	Value in the code	Unit
$l_1,\ l_2$	link lengths (joint to joint)	0.5, 0.4	m

$m_1,\ m_2$	link masses (without payload)	2.0, 1.5	kg
$l_{c1},\ l_{c2}$	distance from a link's joint to its own centre of mass (COM)	0.25, 0.2 (middle of each link)	m
$I_1,\ I_2$	moment of inertia of a link about its own COM	$ml^2/12$ (uniform rod)	kg m^2
g	gravitational acceleration	9.81	m/s^2
m_p	point-mass payload at the tip of link 2 (varies per sample)	0 to 1 (dataset), up to 1.5 out-of-range	kg
b	viscous joint friction coefficient (varies per sample)	0.05 to 0.5	N m s/rad
$q_1,\ q_2$	joint angles: q_1 from the horizontal, q_2 relative to link 1	state	rad
$\dot{q}_1,\ \dot{q}_2$	joint angular velocities (a dot means d/dt)	state	rad/s
$\tau_1,\ \tau_2$	motor torques applied at joints 1 and 2 (the inputs)	$A\sin(2\pi f t+\phi)$	N m

Toolbox: Lagrangian mechanics in five minutes

You know Newton's second law for a straight-line motion: $F = ma$. For something that *rotates* about a fixed pin the same law reads

$$\tau = I\,\ddot{\theta},$$

where τ is torque (a twisting push, N m), I the moment of inertia (resistance to being spun, kg m^2) and $\ddot{\theta}$ the angular acceleration. For *one* link that is easy. For *two* links, the second link is moved by the first, which is moved by the base, and the joints push on each other with forces we do not know. Newton's method forces us to compute all those internal forces. **Lagrange's trick** avoids them completely by working with *energies*, which are plain numbers.

The recipe

1. Choose coordinates that describe the pose with no redundancy: here q_1, q_2 (the *generalised coordinates*).
2. Compute the kinetic energy T (as a function of $q, \dot{q}$) and the potential energy V (as a function of q).
3. For every coordinate q_k evaluate the **Euler-Lagrange equation**

$$\underbrace{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right)}_{\text{(a) rate of change of 'momentum'}} - \underbrace{\frac{\partial T}{\partial q_k}}_{\text{(b) pose-dependence of } T} + \underbrace{\frac{\partial V}{\partial q_k}}_{\text{(c) gravity pulls}} = Q_k$$

   where Q_k collects everything we push with from outside (motor torque τ_k, friction).

Reading it: (a) is "how fast is the generalised momentum $\partial T/\partial \dot{q}_k$ changing" (it is the analogue of $d(mv)/dt$); (c) is the force that gravity exerts along coordinate q_k, namely minus the slope of the potential energy (a ball rolls downhill, i.e. the force is $-\partial V/\partial q$, so moving it to the other side of the equation gives $+\partial V/\partial q$); (b) is a correction that appears because the "mass" itself can depend on the pose.

The symbol ∂ means a **partial derivative**: differentiate with respect to one variable while pretending all the others are constants. For instance if $T = \tfrac{1}{2}(3+\cos q_2)\,\dot{q}_1^2$ then $\partial T/\partial \dot{q}_1 = (3+\cos q_2)\dot{q}_1$ and $\partial T/\partial q_2 = -\tfrac{1}{2}\sin q_2\,\dot{q}_1^2$.

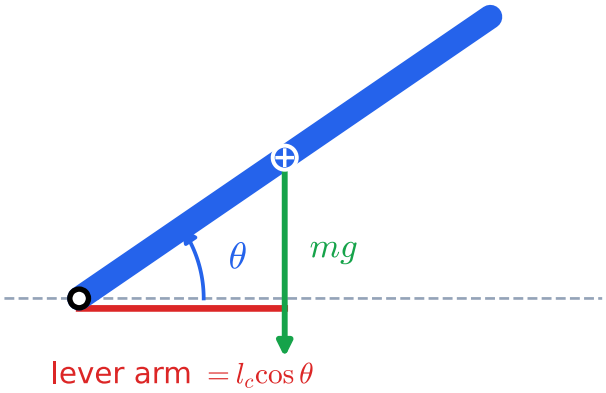
Test run: one swinging rod (you can check it with Newton)

A uniform rod of mass m pivots at the origin, angle θ from the horizontal; its COM is a distance l_c from the pivot. Its inertia about the pivot is $I_p = I_{\text{cm}} + ml_c^2$ (explained in the *combine_payload* document). Then

$$T = \tfrac{1}{2} I_p \dot{\theta}^2, \qquad V = mg\, l_c \sin\theta.$$

Euler-Lagrange: $\frac{d}{dt}(I_p\dot{\theta}) - 0 + mgl_c\cos\theta = \tau$, i.e.

$$I_p\ddot{\theta} + mg\, l_c \cos\theta = \tau.$$



Newton's view of the same rod. Gravity pulls down with force mg at the COM. Only the horizontal distance $l_c\cos\theta$ to the pivot gives a twist (the lever arm), so gravity's torque about the pin is $mg\, l_c \cos\theta$ - exactly the term Lagrange produced.

Newton gives $I_p\ddot{\theta} = \tau - (\text{gravity torque})$, the same equation. So the method works, and for two links it needs only energies. That is the whole idea; the rest is careful algebra.

1 The question this function answers

Hold your arm out horizontally and you feel it: gravity wants to pull it down, and your muscles must supply a steady counter-torque even though nothing is moving. The robot is the same. `gravity_vector` returns the pair of joint torques $G(q) = (G_1, G_2)$ that exactly balance gravity when the arm is held still at pose q. It is an *arm-sized version of "weight times lever arm"*, and it depends only on the pose, not on speed.

```
def gravity_vector(q, p):
    q1, q2 = q
    m2e, lc2e, I2e = combine_payload(p.m2, p.lc2, p.I2, p.l2, p.m_p)
    g1 = (p.m1*p.lc1 + m2e*p.l1) * p.g * np.cos(q1) + m2e*lc2e*p.g*np.cos(q1 + q2)
    g2 = m2e*lc2e*p.g*np.cos(q1 + q2)
    return np.array([g1, g2])
```

We derive it twice, in two completely different ways, which must agree: first from **energy** (the Lagrange route), then from **lever arms** (the Newton route).

2 Route A: from potential energy

Step 1: heights

Gravitational potential energy of a mass m at height y is $V = mgy$ (the work you do lifting it). We measure heights from joint 1 ($y = 0$ at the base). By trigonometry the heights of the two centres of mass are

$$y_1 = l_{c1}\sin q_1, \qquad y_2 = l_1 \sin q_1 + l_{c2e}\sin(q_1 + q_2)$$

(elbow height plus the vertical rise along link 2, whose direction is $q_1 + q_2$). A rigid body's gravitational energy equals Mgy_{cm} (see *combine_payload*), so with the payload folded into link 2:

$$V = g\Big[m_1 l_{c1}\sin q_1 + m_{2e}\big(l_1 \sin q_1 + l_{c2e}\sin(q_1+q_2)\big)\Big].$$

Step 2: gravity's generalised force

In the Euler-Lagrange equation gravity appears as $\partial V/\partial q_k$. Why that is the holding torque: if joint k turns by a small angle dq_k, the energy stored in height changes by $dV = \frac{\partial V}{\partial q_k} dq_k$. Moving slowly, the motor does work $\tau_k\, dq_k$, and that work must all go into potential energy, so $\tau_k = \partial V/\partial q_k$. (The same result drops out of the Euler-Lagrange equation with $\dot q = \ddot q = 0$: all kinetic terms vanish and $\partial V/\partial q_k = \tau_k$ remains.)

Step 3: differentiate

Remember $\frac{d}{dx}\sin x = \cos x$ and, by the chain rule, $\frac{\partial}{\partial q_1}\sin(q_1+q_2) = \cos(q_1+q_2)\cdot 1$ (the same for q_2).

$$G_1 = \frac{\partial V}{\partial q_1} = g\Big[m_1 l_{c1}\cos q_1 + m_{2e}\big(l_1\cos q_1 + l_{c2e}\cos(q_1+q_2)\big)\Big]$$

$$G_2 = \frac{\partial V}{\partial q_2} = g\, m_{2e}\, l_{c2e}\cos(q_1+q_2)$$

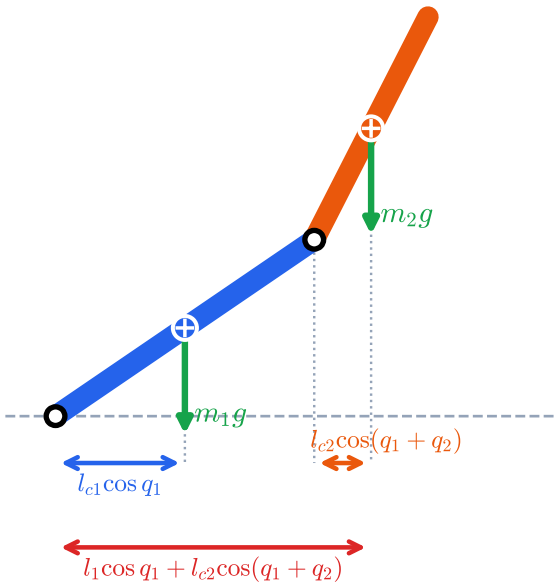
Regroup the $\cos q_1$ terms in G_1 and you have the code:

$$\boxed{\begin{aligned} G_1 &= \big(m_1 l_{c1} + m_{2e} l_1\big)\, g\cos q_1 + m_{2e} l_{c2e}\, g\cos(q_1+q_2) \\ G_2 &= m_{2e}\, l_{c2e}\, g\cos(q_1+q_2) \end{aligned}}$$

3 Route B: lever arms (pure Newton) — the same answer

A force produces a torque about a pivot equal to **force** $\times$ **horizontal distance from the pivot** (for a vertical force). The "horizontal distance" is the lever arm; the vertical component of position is irrelevant.

- **Joint 2** has to hold up link 2 (+payload), whose weight $m_{2e}g$ acts at COM 2. That COM is a horizontal distance $l_{c2e}\cos(q_1+q_2)$ from the elbow (link 2 points at angle q_1+q_2). So $G_2 = m_{2e}g\, l_{c2e}\cos(q_1+q_2)$. ✓
- **Joint 1** has to hold up *everything*. Weight of link 1, $m_1 g$, has lever arm $l_{c1}\cos q_1$. Weight of link 2, $m_{2e}g$, has lever arm = horizontal distance from the base to COM 2 $= l_1\cos q_1 + l_{c2e}\cos(q_1+q_2)$. Add: $G_1 = m_1 g\, l_{c1}\cos q_1 + m_{2e}g\,\big[l_1\cos q_1 + l_{c2e}\cos(q_1+q_2)\big]$. ✓ identical to Route A.

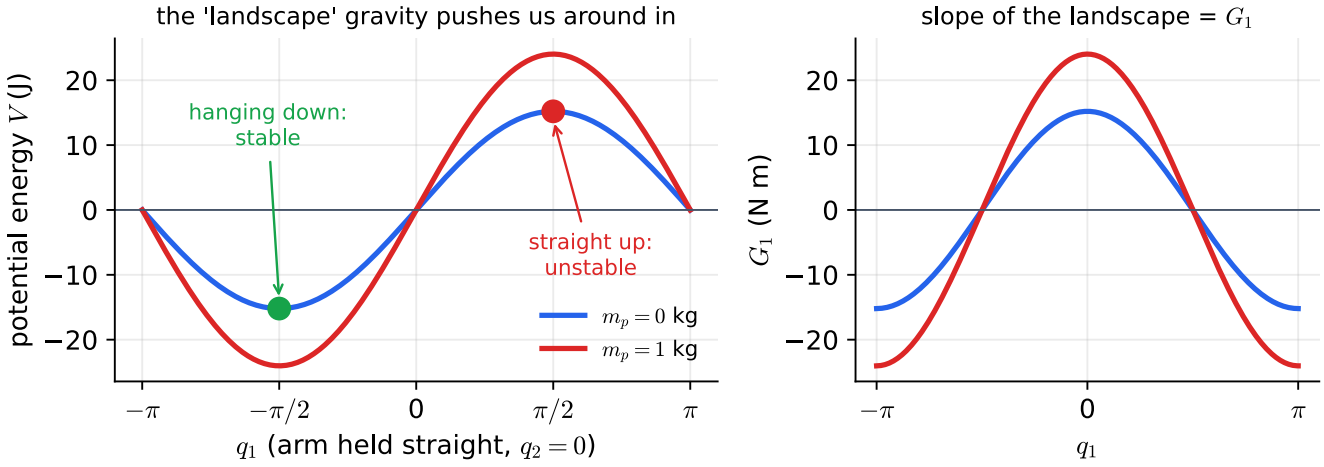


Gravity as lever arms. Green: the weights acting at each centre of mass. Dimension lines along the bottom are the horizontal lever arms. Joint 1 must resist the weight of link 1 (blue lever arm) *plus* the weight of link 2 (red lever arm, which spans both links); joint 2 only sees link 2 (orange lever arm).

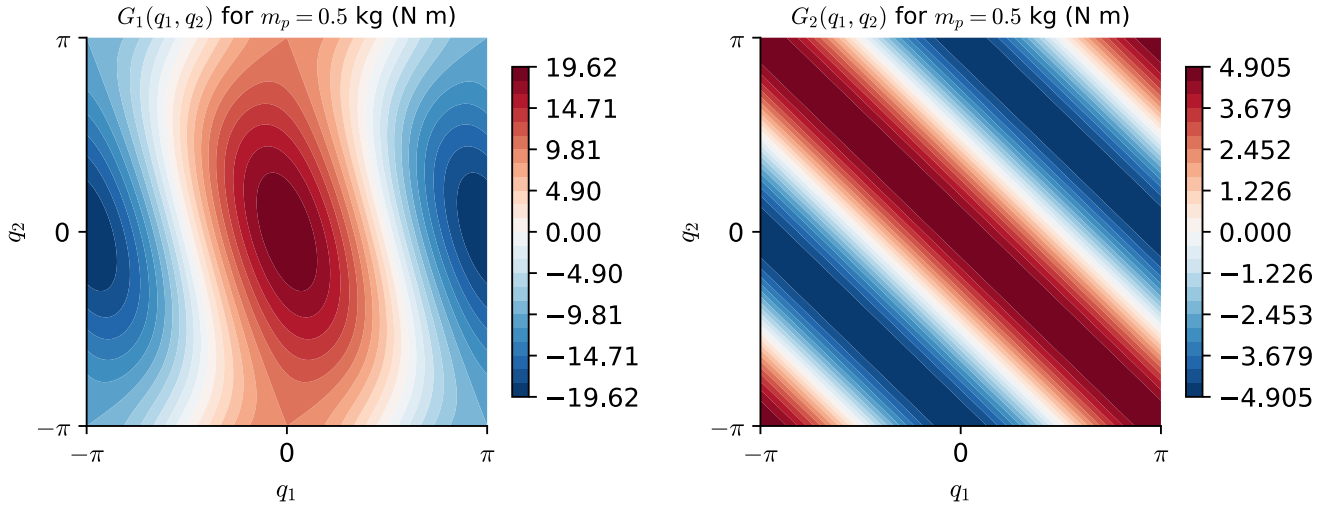
> **INTUITION**
> Every term has the form **(weight)** $\times$ **(horizontal distance)**, so it is largest when the arm is horizontal ($\cos = 1$) and zero when it points straight up or down ($\cos = 0$, the weight acts right along the arm and has no lever). The payload simply adds $m_p g\times$ its lever arm, because $m_{2e} l_{c2e} = m_2 l_{c2} + m_p l_2$.

4 Reading the formula

- **Sign convention.** G is the torque the motor must supply, in the same direction as positive q (anticlockwise). Gravity pulls a horizontal arm down (clockwise), so holding it needs *positive* torque: $G_1 > 0$ at $q_1 = 0$. ✓
- **Joint 2 knows only the absolute angle** $q_1 + q_2$ of link 2, because the weight acts on the link in the real world, not relative to link 1.
- G **does not depend on velocity** and not on friction or inertia.
- **Equilibria.** If $G = 0$ and no torque is applied, the arm stays put. $G_2 = 0$ needs link 2 vertical; $G_1 = 0$ then needs link 1 vertical as well (or the two contributions cancelling). Hanging straight down ($q = (-\pi/2, 0)$) and standing straight up ($q = (\pi/2, 0)$) are both equilibria; the project's unit test `test_hanging_straight_down_is_equilibrium` checks the first. Down is **stable** (valley), up is **unstable** (hilltop), see the plot.



Left: the potential energy $V(q_1)$ with the arm held straight. Think of a ball rolling on this curve: it settles in the valley at $-\pi/2$ and would balance (precariously) on the peak at $+\pi/2$. **Right:** G_1 is exactly the *slope* of that curve, which is why $G = \partial V / \partial q$. A heavier payload scales the whole landscape up.



Gravity torque for every pose (payload 0.5 kg). Red = needs positive (anticlockwise) torque, blue = negative. The patterns are cosines of q_1 and of $q_1 + q_2$. G_2 depends on $q_1 + q_2$ only, hence its diagonal stripes.

5 Worked example with real numbers

$m_p = 0.5$ kg, so $m_{2e} = 2.000$, $l_{c2e} = 0.250$. Pose $q = (0.5,\ 0.8)$: $\cos q_1 = 0.8776$, $\cos(q_1 + q_2) = \cos 1.3 = 0.2675$.

quantity	calculation	value
$(m_1 l_{c1} + m_{2e} l_1) g$	$(2.0 \cdot 0.25 + 2.000 \cdot 0.5) \cdot 9.81 = 1.500 \cdot 9.81$	14.7150
$m_{2e} l_{c2e} g$	$2.000 \cdot 0.250 \cdot 9.81$	4.9050
G_1	$14.7150 \cdot 0.8776 + 4.9050 \cdot 0.2675$	14.2257 N m
G_2	$4.9050 \cdot 0.2675$	1.3121 N m

Exactly the output of `gravity_vector(q, p)`. Biggest case (arm horizontal, $q = 0$): $G = (15.21,\ 2.94)$ N m with no payload and $(24.03,\ 6.87)$ N m with a 1 kg payload. Compare with the motor torque amplitudes of up to 15 N m in the dataset: the motors can compete with gravity but the payload makes it close.

6 Code ↔ maths dictionary

code	maths
`(p.m1*p.lc1 + m2e*p.l1) * p.g * np.cos(q1)`	$(m_1 l_{c1} + m_{2e} l_1)\, g \cos q_1$: both links' weights acting through the $\cos q_1$ lever arms of link 1
`m2e*lc2e*p.g*np.cos(q1 + q2)`	$m_{2e} l_{c2e} g \cos(q_1 + q_2)$: weight of link 2 through its own lever arm
`g2 = ...`	$G_2 = \partial V / \partial q_2$

CHECK YOURSELF

1. Arm straight up, $q = (\pi/2, 0)$: what is G? *Answer:* $\cos(\pi/2) = 0$ in both terms, so $G = (0, 0)$.
2. Arm horizontal with the elbow bent straight up ($q = (0, \pi/2)$): G_2? *Answer:* $\cos(\pi/2) = 0$, so link 2 hangs vertically above the elbow and needs no torque; $G_1 = (m_1 l_{c1} + m_{2e} l_1) g$ (the weight of link 2 acts at horizontal distance l_1).
3. With the arm horizontal ($q = 0$), by how much does a payload m_p raise G_1 and G_2? *Answer:* G_1 rises by $m_p g (l_1 + l_2)$ and G_2 by $m_p g\, l_2$: the payload's lever arm is the distance from the pivot to the tip.